



PAPER SUBMISSION DEADLINE: JULY 1, 2026



AgenticOS 2026

The 2nd Workshop on Operating Systems Design for AI Agents



Congress Hotel Prague, Czechia | 17 Sept 29, 2026 | Co-located with SOSP 2026

THE CHALLENGE

THE LEGACY OS IS HOLDING AI BACK.

AI agents have outgrown their sandboxes. Today's autonomous agents plan, invoke tools, and collaborate continuously. Yet, they are bottlenecked by decades-old OS abstractions—processes, threads, and sockets—that were never built for dynamic, semantically rich AI workloads.

To unleash AI at scale, the OS itself must become agentic. We must rebuild from the kernel up: rethinking isolation, semantic scheduling, and context management.

CORE THEMES & TOPICS

- **New Abstractions:** Process, container, and multikernel enhancements tailored for agent execution.
- **Dynamic Sandboxing:** Lightweight, iron-clad runtimes for executing unverified, agent-generated code.
- **Semantic Scheduling:** Resource management aware of multi-agent and LLM prompt workloads.
- **Episodic Memory:** Long-lived state abstractions managing context and prompt histories natively.
- **Agent Observability:** Deep provenance tracking and debugging for autonomous operations.
- **Compiler-OS Co-design:** Adaptive, agent-aware JIT execution strategies.
- **Self-Managing Systems:** Agents taking the wheel for kernel tuning, anomaly detection, and failure recovery.

CALL FOR PAPERS

We are actively seeking pioneers, researchers, and open-source practitioners to share position papers, real-world experience reports, and breakthrough research. (*ACM double-column format*).

TRACK 1: VISION PAPERS (1-2 PAGES)

Spark the debate. Pitch early-stage ideas, bold position statements, ongoing projects, demos, or critical insights from production systems.

TRACK 2: RESEARCH PAPERS (UP TO 6 PAGES)

Define the science. Submit complete architectural concepts, rigorous research results, and detailed experience reports.

CRITICAL DATES

Paper Submission	July 1, 2026
Acceptance Notice	August 7, 2026
Camera-Ready	August 29, 2026
Workshop Date	September 29, 2026

THE ORGANIZERS

Yubin Xia & Dong Du — Shanghai Jiao Tong Univ.

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JOIN THE REVOLUTION. SUBMIT TODAY.

Scan the QR code to read the full CFP and submit your paper via HotCRP.

<https://os-for-agent.github.io/>

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